

Retrieval Evaluation Report

Evaluated at: 2026-07-18T05:49:20.303Z
Total Queries: 65 | K = 5

Method	P@5	R@5	MRR	NDCG@5	Latency(ms)
FAISS	0.2185	0.6615	0.5992	0.5922	9
BM25	0.3169	0.9064	0.8800	0.8782	125
HNSW	0.2185	0.6615	0.5992	0.5922	144
Hybrid (FAISS+BM25)	0.2892	0.8538	0.8054	0.7803	137
Rerank (FAISS+Cohere)	0.2185	0.6692	0.6615	0.6159	755
GraphRAG	0.1723	0.5590	0.6985	0.5629	1670

Winners by Metric

Best Precision@5: BM25 (0.3169)
Best Recall@5: BM25 (0.9064)
Best MRR: BM25 (0.8800)
Best NDCG@5: BM25 (0.8782)
Fastest: FAISS (9 ms avg)

Methodology Note: Rerank (FAISS+Cohere) latency figures exclude a 6-second per-query sleep introduced to comply with the Cohere free-tier rate limit (10 req/min). This sleep is a deployment constraint, not a retrieval cost. Actual wall-clock time per Rerank query is ~6 seconds higher than the value shown. All other methods report unmodified wall-clock latency.

Statistical Significance Analysis

Comparing Best (BM25) vs Second Best (Hybrid (FAISS+BM25)) using Wilcoxon Signed-Rank Test:
p-value: 0.0019 (Statistically Significant at alpha=0.05)
This confirms whether the top method's performance advantage is scientifically robust.